Prestudy - Calibrating a Helmholtz coil and function tests of cryogenic PCB

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Bachelor's thesis project -

A cryogenic magnetic field sensor for quantum computing

1 Introduction

This prestudy presents the background theory and methodology for calibrating a constructed Helmholtz coil. Furthermore, the coil will be used to perform a function test of the hall - element to ensure functionality for later tests.

1.1 The Helmholtz coil

A Helmholtz coil is a pair of coils with a certaing geometry which makes it ideal for creating a homogenous magnetic field with a highly controllable magnetic flux density. A schematic picture of the coil is shown in figure 1.

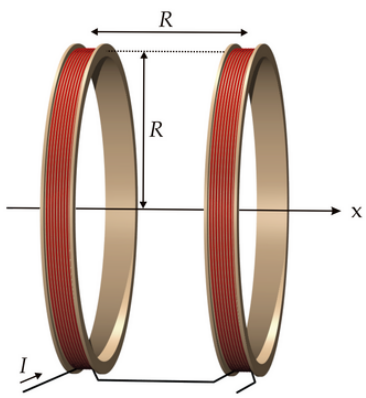

Figure 1: *Schematic figure for a helmholtz coil. The setup consists of two coils of equal geometry. By placing the coils the same distance apart as their individual geometry, a homogenous magnetic field is created in the center. By running the same current through both coils, the magnetic flux density can be easily computed.*

By placing the coil with a distence apart that is the same as their individual radius, the magnetic flux density in the center of the coils can be calculated as:

$$B = \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 N I}{R}, \tag{1}$$

with μ_0 being the magnetic permeability, N the number of turns per coil, I the current and R the radius.

The group has already counstructed a helmholtz coil. For this coil, the radius is ≈ 3 cm and each coil has aroun 100 turns (we might have lost count at some point during the winding). The coil has been verified to conduct current and has a resistance of around 2 Ω each. Using the geometry of our coil and expressing the magnetic flux density B as a function of the current I, the expression would be as follows:

$$\left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 N}{R} = 2.6 \cdot 10^{-3} \rightarrow B = 2.6\, I \cdot 10^{-3} \tag{2}$$

We should therefore expect to see approximately 2.6 mT/A in the center of our coil.

1.2 The Hall element

To measure the magnetic field, a commercial Hall effect sensor is used. The sensor is the HQ0811 from AKM. It has been prepared for calibration by being mounted to a small pcb, as shown in figure 2.

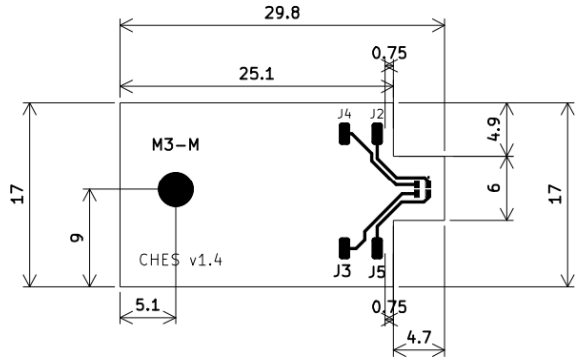

Figure 2: *Technical drawing of the cryogenic PCB. The Hall element is placed furthest to the right.*

By providing a current through the hall element, a measureable Hall voltage should be produced. The formuala for the hall-voltage is given as

$$V_H = \frac{BI}{dnq} \tag{3}$$

with the magnetic flux denisty being B, the current I. d is the thickness of the hall element, q is the charge of the charge carrier and n is the charge carrier density. Since d, q and n all are constant (n changes as a function of temperature, and we can assume the temperature is constant), the hall voltage should be linearly proportional to both the magnetic flux density and the current.

Reading the datasheet for the Hall element, the produced Hall voltage is a linear function of the magnetic flux density of approximately

$$V_H = 3.15B \tag{4}$$

with a constant current of 5 mA.

Thus, by providing a steady current to our hall element we should be able to calibrate the hall voltage to the changing magnetic flux density, which we can control through our Helmholtz coil.

2 Methodology

This section briefly summarizes the needed equipment and how the experiments are planned to be carried out.

2.1 Equipment

To perform these tests, we need to place the coil in a magnetically shielded environment. We thus need a cryopermic shield to shield out other magnetic fields. We also need an external gaussmeter to calibrate the coil.

To provide a steady current through the coil and the hall element we need at least 2 DC power supplys with controllable current. To measure the hall voltage we also need a multimeter. We also need access to some construction tools, mainly wires and soldering equipment.

The equipment needed is summarized in the following list:

- Gaussmeter (external, shown to us by Robert a few weeks back).
- Cryopermic shield.
- 2 DC power supplys for providing current through coil and hall element.

- Multimeter for measuring Hall voltage.
- Misc. tools such as wires, soldering equipment etc (might not be needed but very good to have in case of an emergency).

2.2 Experiments

Initially, the coil will be placed inside the cryopermic shield and the background magnetic flux density will be measured. Then, a controlled current will be applied to the coil and the produced magnetic flux denisty will be measured. This measurement will be performed for several currents over several orders of magnitude.

When calibration of the coil is done, the Hall element PCB will be placed inside the cryopermic shield and a constant magnetic field will be provided to it. A constant current of 5 mA will be applied to the hall element and the resulting Hall voltage will be measured.

3 Expected results

We expect to be able to produce a magnetic field at the center of the Helmholz coil linearly proportional to the current applied over the copper wire. This relationship should be approximately 2.6 mT/A, but matching this relationship is not crucial. As long as a linear relationship between the current and magnetic flux density can be established, the test will be a success.

Regarding the Hall element, calibrating it is not the main purpose of this test. The purpose of this test is to establish functionality, so that we with good confidence can move further in this project and later test the hall element with the full measurement setup. The full measurement setup will also contain the room-temp PCB for data aquisition. The test of the cryo PCB will be considered a success if the measured Hall voltage is found to be linearly dependant on the applied magnetic field or current.

The produced hall voltage will however be quite small, thus needing amplification in the test stage. This extra amplification further complicates calibration, since the amplification in the initial test stage won't be the same as the amplification of the final product. It will however not interfere with function tests.